Memory, Vault, CASS, and Smart Memory MCP

Standalone technical deep-dive dossier

Memory, Vault, CASS, and Smart Memory MCP

Long-running agents need more than a large context window. They need raw memory, compressed memory, procedural learning, contradiction handling, retrieval, and a way to avoid acting on stale facts.

- This shows that long-running agents need memory governance: raw records, summaries, procedural learning, contradictions, staleness, and retrieval metadata.
- Presented as practical memory infrastructure, not proof of human-like consciousness.

What I had in mind

The memory work came from noticing that agents do not only need more context. They need memory hygiene: raw facts, summaries, procedural learning, contradictions, staleness and provenance.

What I built

- `cass_pipeline.py`, `memory_broker.py`, vault-watcher design notes, and `smart-memory-mcp/server.js` show schemas, scoring, contradiction logic, TF-IDF retrieval, recency/usefulness metadata, and MCP tool declarations.

Mechanism detail

- The memory work is strongest when framed as reliability infrastructure. It separates raw episodic records from compressed working memory and procedural lessons.
- The broker/broadcast layer is a response to module isolation: important facts, central notes and contradictions need to move between subsystems rather than stay trapped.
- The vault watcher and Smart Memory MCP ideas make memory more than a folder. They introduce indexing, recency, usefulness, access metadata, graph context and stale-information handling.

Architecture

- CASS keeps episodic raw records, structured working memory, and procedural rules with confidence/decay.
- Memory Broker broadcasts high-value or central facts across memory systems and flags contradictions.
- Vault watcher design handles file-change detection, incremental reindexing, graph updates, staleness propagation, and forgetting scores.
- Smart Memory MCP provides a simple local protocol component for learn/recall/stats/patterns/suggestions.

Evidence cluster

- Memory Broker source showing broadcast schema, write logs, weight/centrality scoring, and lightweight contradiction detection.
- CASS source implementing episodic raw memory, working summaries, procedural rules, confidence, decay, and SQLite persistence.
- Vault watcher design notes covering file-change detection, incremental indexing, graph updates, staleness propagation, and forgetting score.
- Smart Memory MCP source exposing learn/recall/stats/pattern/suggestion tools with retrieval metadata.

Claim-to-evidence map

The public version should be strong because it is bounded, not because it overclaims.

Claim	Evidence	Boundary
This shows that long-running agents need memory governance: raw records, summaries, procedural learning, contradictions, staleness, and retrieval metadata.	Memory Broker source showing broadcast schema, write logs, weight/centrality scoring, and lightweight contradiction detection.; CASS source implementing episodic raw memory, working summaries, procedural rules, confidence, decay, and SQLite persistence.	Presented as practical memory infrastructure, not proof of human-like consciousness.
The work is valuable because it shows mechanism, not surface polish.	CASS keeps episodic raw records, structured working memory, and procedural rules with confidence/decay.; Memory Broker broadcasts high-value or central facts across memory systems and flags contradictions.	Fresh public demos or benchmarks would be the next proof layer.

Senior-level signal

This shows that long-running agents need memory governance: raw records, summaries, procedural learning, contradictions, staleness, and retrieval metadata.

- Context integrity: summaries are useful but dangerous when they replace raw evidence.
- Contradiction handling: conflicting facts are not simply overwritten.
- Agent safety relevance: stale memory can cause wrong action, wrong confidence and wrong long-term behavior.

Skeptical reviewer questions

- What is actually built here, and what is still design? Answer: the status and boundary are stated directly inside the Memory, Vault, CASS, and Smart Memory MCP case.
- Which private evidence is intentionally not public? Answer: local paths, credentials, private channel details and confidential raw material are excluded.
- Why is this relevant? Answer: the case shows a reusable component for agents, tool use, memory, routing, validation or high-stakes governance.
- What would be the next stronger proof? Answer: synthetic demo, audit trace, benchmark or external review, depending on the case.

Lessons and boundaries

Presented as practical memory infrastructure, not proof of human-like consciousness.

- Agents can become dangerous or useless when they act on stale summaries.
- Raw memory should remain available when summaries become distorted.
- Contradictions should be preserved and surfaced instead of overwritten.
- Retrieval scoring needs more than text similarity; usefulness and recency matter.

Next hardening / next project step

- Add a public synthetic-memory demo.
- Benchmark retrieval and contradiction detection on non-private data.
- Expose memory provenance in every agent response.
- Add explicit redaction rules for private vault content.
- Create a synthetic vault with deliberate contradictions and stale facts.
- Show how the agent retrieves raw episodes, working summaries and procedural rules with provenance.
- Benchmark retrieval quality and contradiction surfacing on public data only.